IoT Weather Station Telemetry Platform

Real time atmospheric monitoring with ESP32 microcontrollers, cloud MQTT ingestion, embedded time series storage and reactive web dashboards

THE PROBLEM

Environmental telemetry is either costly, closed proprietary hardware or slow batch loggers with high latency.

OUR SOLUTION

An open, production ready IoT station: low-cost sensors stream live microclimate data through a decoupled MQTT broker to embedded storage and a real time web dashboard.

UNDER THE GUIDANCE OF

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TEAM MEMBERS

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How we achieved the project

1 Hardware & signal conditioning

- ESP32 DevKit V1: dual core Xtensa LX6 @ 240 MHz
- DHT22 (temp, humidity) + BMP180 (pressure)
- Breakthrough: an interrupt safe µs pulse reader (noInterrupts) with internal pullups eliminated sensor TIMEOUT errors under heavy campus network traffic

2 Decoupled MQTT transport

- Moved from point-to-point HTTP to publish subscribe MQTT (port 1883) on HiveMQ (broker.hivemq.com)
- Hierarchical topics: weather/<device_id>/readings
- Last Will & Testament on …/status gives instant hardware offline detection

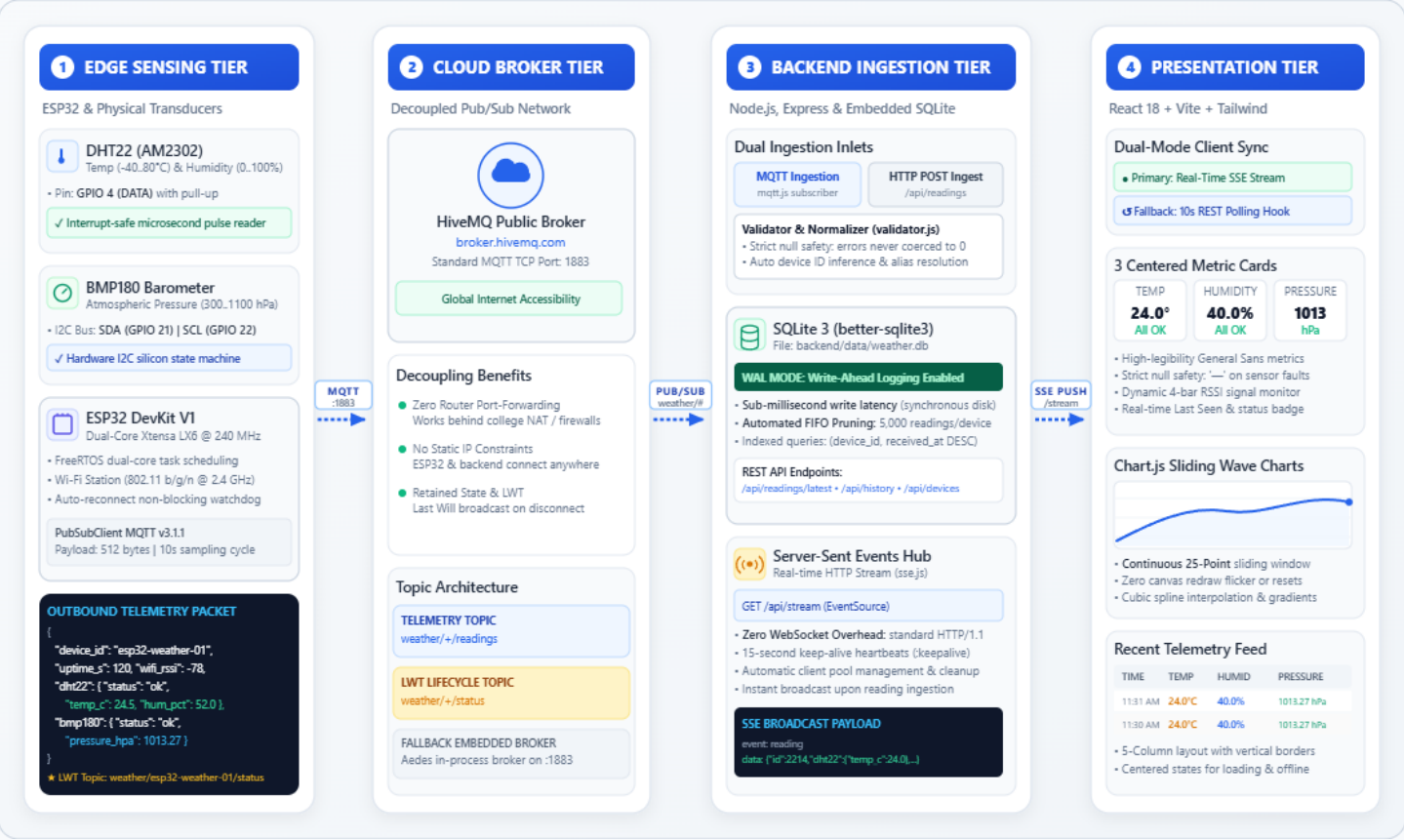
3 Node.js backend & storage

- Express engine with dual ingestion: MQTT subscriber + HTTP REST fallback
- SQLite in WAL mode, FIFO pruning keeps the newest 5,000 readings per device
- SSE hub at /api/stream with 15 s keep alive heartbeats

4 Reactive minimalist dashboard

- React 18, Vite and Tailwind CSS with a light design system and General Sans type
- Chart.js sliding window charts with gradient fills, no canvas flicker
- Live SSE feed with 10 s REST polling fallback

System architecture: four functional tiers



1 Edge tier

Dual sensors: µs bit banged DHT22 pulse train and hardware I2C for the BMP180.

2 Broker tier

Decoupled cloud pub/sub network: no router NAT or port forwarding constraints.

3 Backend tier

Express + SQLite engine with submillisecond insert times.

4 Presentation tier

Browser native EventSource with zero WebSocket handshake overhead.

Unique selling points & technical highlights

1. **Zero cloud cost, globally decoupled mesh**
   Public MQTT broker plus embedded SQLite: no AWS IoT Core, DynamoDB or external database fees.

2. **Strict hardware to UI null safety**
   Sensor faults stay null, never a fake 0 °C or 0 %: the UI shows an empty value and a Degraded status pill.

3. **Self healing dual mode resiliency**
   Firmware automatically reconnects WiFi and MQTT; the dashboard fails over from SSE to 10 s REST polling.

4. **High performance sliding window charts**
   Chart.js slides a 25-reading window with spline interpolation and gradient fills, no flicker.

5. **Industrial minimalist UI & UX**
   Pure light mode, centered type, equal 20 % columns and a live 4 bar RSSI signal indicator.



Future scope & what can be improved

1	Edge ML & barometric forecasting	A Zambretti algorithm or TinyML regression predicts rain, storm fronts and cloud cover 6 to 12 h ahead from the pressure rate of change (ΔP/Δt).
2	Deep sleep & solar harvesting	esp_deep_sleep_start() with ULP wake up every 10 to 15 min cuts draw from ~80 mA to ~10 µA for multiyear off grid operation.
3	Enterprise security: MQTTS & tokens	MQTTS on port 8883 (TLS, X.509 certificates) plus JWT/HMAC payload signing to stop unauthorized injection on public topics.
4	Multinode geospatial mapping	Leaflet or Mapbox tile maps to track distributed stations across campuses, farms and smart city districts.
5	Proactive alerting & webhooks	Web Push, Telegram or Discord alerts for heatwaves, sudden barometric drops and prolonged sensor dropouts.

Thank you

Explore the project

LIVE DASHBOARD
iot-weather-8.vercel.app

SOURCE CODE
github.com/Pranjal-Sahu21/iot-project

Project at a glance

SENSORS
DHT22 and BMP180

TRANSPORT
MQTT on HiveMQ

BACKEND
Node.js and SQLite

DASHBOARD
React with live updates